INCH-POUND

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MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, ARITHMETIC LOGIC UNITS/FUNCTION GENERATORS, MONOLITHIC SILICON

Inactive for new design after 6 September 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-38535.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, arithmetic logic units/function generators. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3).

1.2 Part or Identifying Number (PIN). The PIN should be in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types should be as follows:

Device type	Circuit
01	Arithmetic logic unit/function generator

1.2.2 Device class. The device class should be the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines should be as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
J	GDIP1-T24 or CDIP2-T24	24	Dual-in-line
K	GDFP2-F24 or CDFP3-F24	24	Flat pack
Z	GDFP7-F24 or CDFP8-F24	24	Flat pack
3	CQCC1-N28	28	Square leadless chip carrier

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, 3990 East Broad St., Columbus, OH 43216-5000, or emailed to bipolar@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings.

Supply voltage	-0.5 V to 7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to 5.5 V dc
Storage temperature range	-65° to +150°C
Maximum power dissipation (P_D) 1/	193 mW dc
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases J, K, Z, and 3	(See MIL-STD-1835)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.7 V
Normalized fanout (each output)	
Logical low level	10 maximum
Logical high level	20 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Copies of these documents are available online at http://assist.daps.dla.mil;quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with MIL-PRF-38535.

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Truth tables and logic equations. The truth tables and logic equations shall be specified on figure 2.

3.3.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-PRF-38535, appendix A).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535, appendix B.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Limits		Unit
			Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V, I_{OH} = -400 µA	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V, I_{OL} = 4.0 mA		0.4	V
Input diode clamp	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C		-1.5	V
Low level input current at M input	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 2/	-0.03	-0.40	mA
Low level input current at $\overline{B}$ input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 2/	-0.09	-1.20	mA
Low level input current at S0 or S3 input	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 2/	-0.12	-1.60	mA
Low level input current at S1 or S2 input	I_{IL4}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 2/ A0, A1, A2, A3 = 5.5 V 3/	-0.12	-1.60	mA
Low level input current at Cn input	I_{IL5}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 3/	-0.15	-2.0	mA
Low level input current at $\overline{A}$ input	I_{IL6}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 2/	-0.09	-1.20	mA
High level input current at M input	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V 3/		20	µA
High level input current at $\overline{A}$ or $\overline{B}$ input	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V 3/		60	µA
High level input current at S0 or S3 input	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V 3/		80	µA
High level input current at S1 or S2 input	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 2.7 V 3/ B1, B1, B2, B3 = 5.5 V		80	µA
High level input current at Cn input	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 2.7 V 2/		100	µA
High level input current at M input	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 3/		100	µA
High level input current at $\overline{A}$ or $\overline{B}$ input	I_{IH7}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 3/		300	µA
High level input current at S0 or S3 input	I_{IH8}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 3/		400	µA
High level input current at S1 or S2 input	I_{IH9}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 3/ $\overline{B}$1, $\overline{B}$1, $\overline{B}$2, $\overline{B}$3 = 5.5 V 3/		400	µA
High level input current at Cn input	I_{IH10}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 2/		500	µA
Short circuit output current at $\overline{G}$ output	I_{OS1}	V_{CC} = 5.5 V, $\overline{S}$3, $\overline{A}$3, $\overline{B}$3 = 5.5 V 3/ 4/	-15	-100	mA
Short circuit output current at Cn+4 output	I_{OS2}	V_{CC} = 5.5 V, 3/ 4/	-15	-100	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Limits		Unit
			Min	Max	
Short circuit output current at $\overline{P}$, $\overline{F}$ 0 thru $\overline{F}$ 3 outputs	I_{OS3}	V_{CC} = 5.5 V, 2/ 4/	-15	-100	mA
Collector cutoff current at A = B output	I_{CEX}	V_{CC} = 4.5 V, $V_{(A = B)}$ = 5.5 V, all input = 4.5 V		100	µA
Supply current	I_{CC1}	V_{CC} = 5.5 V, S0 thru S3, M, and A = 5.5 V		35	mA
	I_{CC2}	V_{CC} = 5.5 V, S0 thru S3 and M = 5.5 V, all others grounded		35	mA
Propagation delay times, high to low level output: 5/, 6/					
$\overline{A}$ i or $\overline{B}$ i to any $\overline{F}$, $\overline{sum}$ mode	t_{PHL1}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	37	ns
$\overline{A}$ i or $\overline{B}$ i to any $\overline{F}$, $\overline{diff}$ mode	t_{PHL2}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	55	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{P}$, $\overline{sum}$ mode	t_{PHL3}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	52	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{P}$, $\overline{diff}$ mode	t_{PHL4}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	57	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{G}$, $\overline{sum}$ mode	t_{PHL5}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	42	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{G}$, $\overline{diff}$ mode	t_{PHL6}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	55	ns
Cn to any $\overline{F}$, $\overline{sum}$ mode	t_{PHL7}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	37	ns
$\overline{A}$ i to $\overline{B}$ i to A = B, $\overline{diff}$ mode	t_{PHL8}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	100	ns
Cn to Cn+4, $\overline{sum}$ mode	t_{PHL9}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	37	ns
Cn to Cn+4, $\overline{diff}$ mode	t_{PHL10}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	37	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{C}$n+4, $\overline{sum}$ mode	t_{PHL11}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	64	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{C}$n+4, $\overline{diff}$ mode	t_{PHL12}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	60	ns
$\overline{A}$ i or $\overline{B}$ i to any $\overline{F}$, logic mode	t_{PHL13}	V_{CC} = 5.0 V, See figure 4 logic mode test table	2	48	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Limits		Unit
			Min	Max	
Propagation delay times, low to high level output: 5/, 6/					
$\overline{A}$ i or $\overline{B}$ i to any $\overline{F}$, $\overline{sum}$ mode	t_{PLH1}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	55	ns
$\overline{A}$ i or $\overline{B}$ i to any $\overline{F}$, $\overline{diff}$ mode	t_{PLH2}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	55	ns
$\overline{A}$ i or $\overline{B}$ i to any $\overline{P}$, $\overline{sum}$ mode	t_{PLH3}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	52	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{P}$, $\overline{diff}$ mode	t_{PLH4}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	52	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{G}$, $\overline{sum}$ mode	t_{PLH5}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	51	ns
$\overline{A}$ i or $\overline{B}$ i to $\overline{G}$, $\overline{diff}$ mode	t_{PLH6}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	55	ns
Cn to any $\overline{F}$, $\overline{sum}$ mode	t_{PLH7}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	46	ns
$\overline{A}$ i to $\overline{B}$ i to A = B, $\overline{diff}$ mode	t_{PLH8}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	76	ns
Cn to Cn+4, $\overline{sum}$ mode	t_{PLH9}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	48	ns
Cn to Cn+4, $\overline{diff}$ mode	t_{PLH10}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	48	ns
$\overline{A}$ i or $\overline{B}$ i to Cn+4, $\overline{sum}$ mode	t_{PLH11}	V_{CC} = 5.0 V, See figure 4 $\overline{sum}$ mode test table	2	64	ns
$\overline{A}$ i or $\overline{B}$ i to Cn+4, $\overline{diff}$ mode	t_{PLH12}	V_{CC} = 5.0 V, See figure 4 $\overline{diff}$ mode test table	2	69	ns
$\overline{A}$ i or $\overline{B}$ i to any $\overline{F}$, logic mode	t_{PLH13}	V_{CC} = 5.0 V, See figure 4 logic mode test table	2	57	ns

1/ Condition of inputs specified in table III.

2/ All unspecified inputs at 5.5 V.

3/ All unspecified inputs at 0 V.

4/ Not more than one output should be shorted at a time.

5/ In $\overline{A}$ i and $\overline{B}$ i i = 0, 1, 2, 3.

6/ C_L = 50 pF ±10%, R_L = 2 kΩ ±5%.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9, 10, 11
Group B electrical test parameters when using the method 5005 QCI option	1, 2, 3, 7, 8 9, 10, 11	N/A
Group C end-point electrical parameters	1, 2, 3, 7, 8 9, 10, 11	1, 2, 3
Group D end-point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

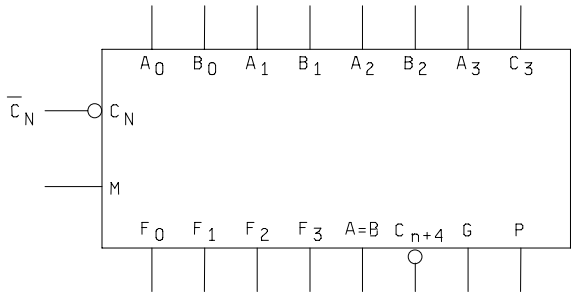
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Terminal number	Terminal symbol	
	Case J, K, Z	Case 3
1	$\overline{B}$ 0	NC
2	$\overline{A}$ 0	$\overline{B}$ 0
3	S3	$\overline{A}$ 0
4	S2	S3
5	S1	S2
6	S0	S1
7	Cn	S0
8	M	NC
9	$\overline{F}$ O	Cn
10	$\overline{F}$ 1	M
11	$\overline{F}$ 2	$\overline{F}$ O
12	GND	$\overline{F}$ 1
13	$\overline{F}$ 3	$\overline{F}$ 2
14	A = B	GND
15	$\overline{P}$	NC
16	Cn+4	$\overline{F}$ 3
17	$\overline{G}$	A = B
18	$\overline{B}$ 3	$\overline{P}$
19	$\overline{A}$ 3	Cn+4
20	$\overline{B}$ 2	$\overline{G}$
21	$\overline{A}$ 2	$\overline{B}$ 3
22	$\overline{B}$ 1	NC
23	$\overline{A}$ 1	$\overline{A}$ 3
24	V_{CC}	$\overline{B}$ 2
25		$\overline{A}$ 2
26		$\overline{B}$ 1
27		$\overline{A}$ 1
28		V_{CC}

Designation	Pin numbers Case J, K, Z	Function
$\overline{A}$ 3, $\overline{A}$ 2, $\overline{A}$ 1, $\overline{A}$ 0	19, 21, 23, 2	Word A inputs
$\overline{B}$ 3, $\overline{B}$ 2, $\overline{B}$ 1, $\overline{B}$ 0	18, 20, 22, 1	Word B inputs
S3, S2, S1, S0	3, 4, 5, 6	Function select inputs
Cn	7	Inv. carry input
M	8	Mode control input
$\overline{F}$ 3, $\overline{F}$ 2, $\overline{F}$ 1, $\overline{F}$ 0	13, 11, 10, 9	Function outputs
A = B	14	Comparator output
$\overline{P}$	15	Carry propagate output
Cn+4	16	Inv. carry output
$\overline{G}$	17	Carry generate output
V_{CC}	24	Supply voltage
GND	12	Ground

FIGURE 1. Terminal connections and pin designations.

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2/	Selection				Active high data 3/		
					M = H	M = L; Arithmetic operations	
	S3	S2	S1	S0	Logic functions	Cn = 0 $\overline{C}$n = 1 = H	Cn = 1 $\overline{C}$n = 0 = H
0	L	L	L	L	F = $\overline{A}$	F = A	F = A plus 1
1	L	L	L	H	F = $\overline{A+B}$	F = A + B	F = (A + B) plus 1
2	L	L	H	L	F = $\overline{A}$ B	F = A + $\overline{B}$	F = (A + $\overline{B}$) plus 1
3	L	L	H	H	F = 0	F = minus 1 (2's compl)	F = zero
4	L	H	L	L	F = $\overline{AB}$	F = A plus A$\overline{B}$	F = A plus A$\overline{B}$ plus 1
5	L	H	L	H	F = $\overline{B}$	F = (A + B) plus A$\overline{B}$	F = (A + B) plus A$\overline{B}$ plus 1
6	L	H	H	L	F = A⊕B	F = A minus B minus 1	F = A minus B
7	L	H	H	H	F = A$\overline{B}$	F = A$\overline{B}$ minus 1	F = A$\overline{B}$
8	H	L	L	L	F = $\overline{A}$ + B	F = A plus AB	F = A plus AB plus 1
9	H	L	L	H	F = $\overline{A⊕B}$	F = A plus B	F = A plus B plus 1
10	H	L	H	L	F = B	F = (A + $\overline{B}$) plus AB	F = (A + $\overline{B}$) plus AB plus 1
11	H	L	H	H	F = AB	F = AB minus 1	F = AB
12	H	H	L	L	F = 1	F = A plus A 1/	F = A plus A plus 1
13	H	H	L	H	F = A + $\overline{B}$	F = (A + B) plus A	F = (A + B) plus A plus 1
14	H	H	H	L	F = A + B	F = (A + $\overline{B}$) plus A	F = (A + $\overline{B}$) plus A plus 1
15	H	H	H	H	F = A	F = A minus 1	F = A

1/ Each bit is shifted to the next more significant position.
2/ This device (ALU) can be used as a comparator when placed in the subtract mode (i. e., S3 S2 S1 S0 are at logical levels LHHL respectively) and the following expressions are valid:

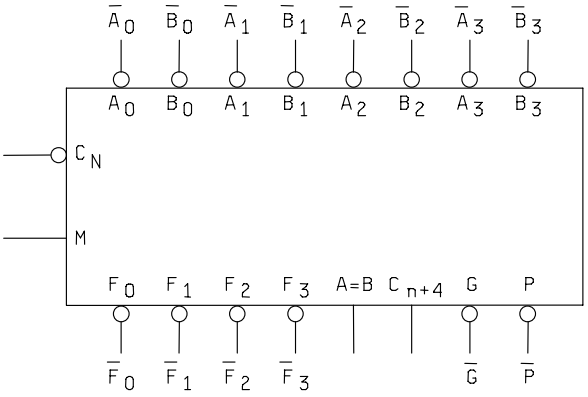
Active high data

When Cn is high and Cn+4 is high, then A ≤ B
When Cn is low and Cn+4 is high, then A < B
When Cn is high and Cn+4 is low, then A > B
When Cn is low and Cn+4 is low, then A ≥ B

3/ The table shown applies for positive logic. If negative logic is used, active high data becomes active low data.

FIGURE 2. Truth tables and logic equations for device type 01.

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2/	Selection				Active low data 3/		
					M = H	M = L; Arithmetic operations	
	S3	S2	S1	S0	Logic functions	Cn = 0 = L	Cn = 1 = H
0	L	L	L	L	$F = \overline{A}$	F = A minus 1	F = A
1	L	L	L	H	$F = \overline{AB}$	F = AB minus 1	F = AB
2	L	L	H	L	$F = \overline{A} + B$	$F = A\overline{B}$ minus 1	$F = A\overline{B}$
3	L	L	H	H	F = 1	F = minus 1 (2's compl)	F = zero
4	L	H	L	L	$F = \overline{A+B}$	$F = A$ plus $(A + \overline{B})$	$F = A$ plus $(A + \overline{B})$ plus 1
5	L	H	L	H	$F = \overline{B}$	$F = AB$ plus $(A + \overline{B})$	$F = AB$ plus $(A + \overline{B})$ plus 1
6	L	H	H	L	$F = \overline{A \oplus B}$	F = A minus B minus 1	F = A minus B
7	L	H	H	H	$F = A + \overline{B}$	$F = A + \overline{B}$	$F = (A + \overline{B})$ plus 1
8	H	L	L	L	$F = \overline{A}\,B$	F = A plus (A + B)	F = A plus (A + B) plus 1
9	H	L	L	H	$F = A \oplus B$	F = A plus B	F = A plus B plus 1
10	H	L	H	L	F = B	$F = A\overline{B}$ plus (A + B)	$F = A\overline{B}$ plus (A + B) plus 1
11	H	L	H	H	F = A + B	F = A + B	F = (A + B) plus 1
12	H	H	L	L	F = 0	F = A plus A 1/	F = A plus A plus 1
13	H	H	L	H	$F = A\overline{B}$	F = AB plus A	F = AB plus A plus 1
14	H	H	H	L	F = AB	$F = A\overline{B}$ plus A	$F = A\overline{B}$ plus A plus 1
15	H	H	H	H	F = A	F = A	F = A plus 1

1/ Each bit is shifted to the next more significant position.
2/ This device (ALU) can be used as a comparator when placed in the subtract mode (i. e., S3 S2 S1 S0 are at logical levels LHHL respectively) and the following expressions are valid:

Active high data

When Cn is low and Cn+4 is low, then $A \leq B$
When Cn is high and Cn+4 is low, then $A < B$
When Cn is low and Cn+4 is high, then $A > B$
When Cn is high and Cn+4 is high, then $A \geq B$

3/ The table shown applies for negative logic. If negative logic is used, active low data becomes active high data.

FIGURE 2. Truth tables and logic equations for device type 01 - Continued.

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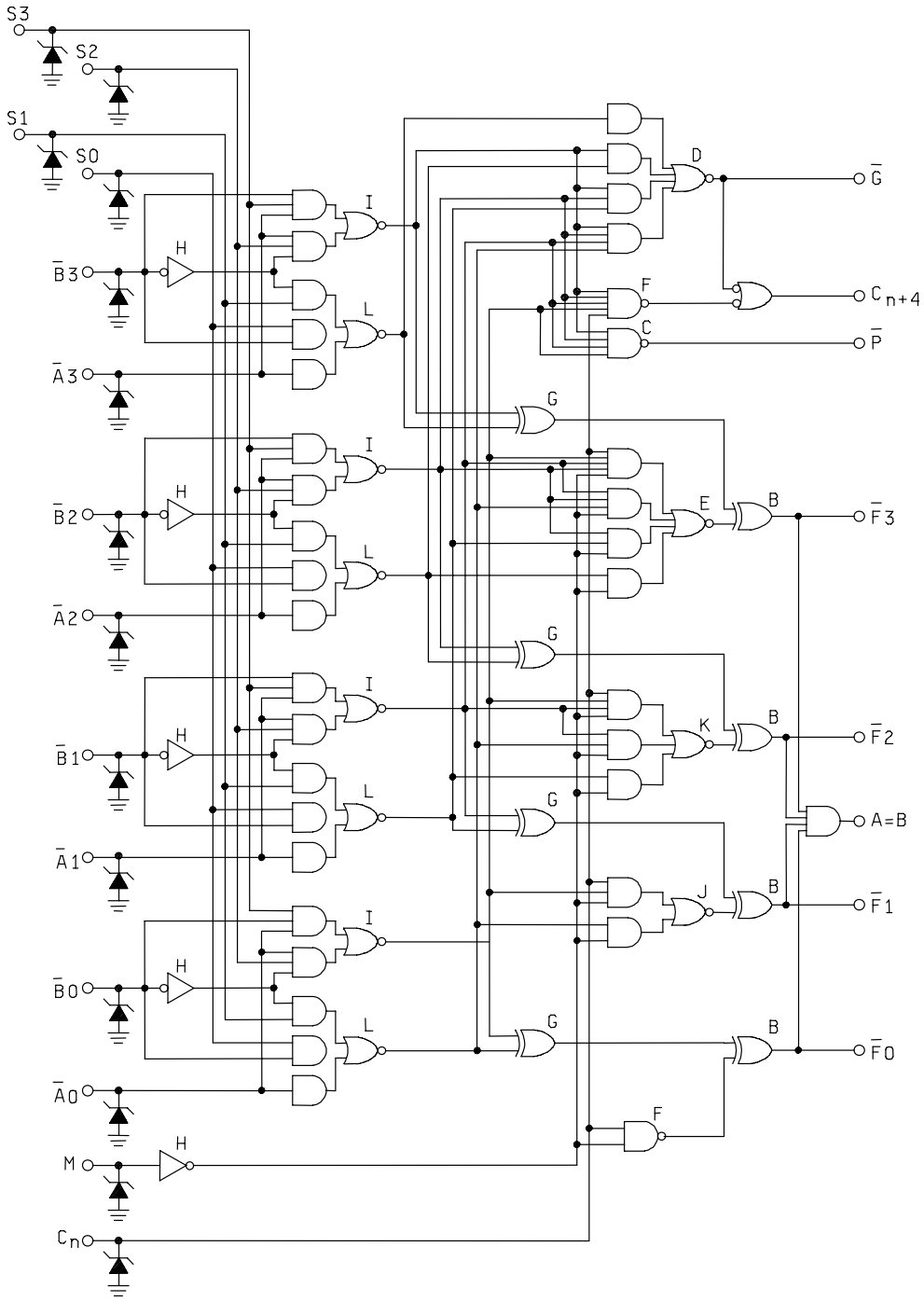
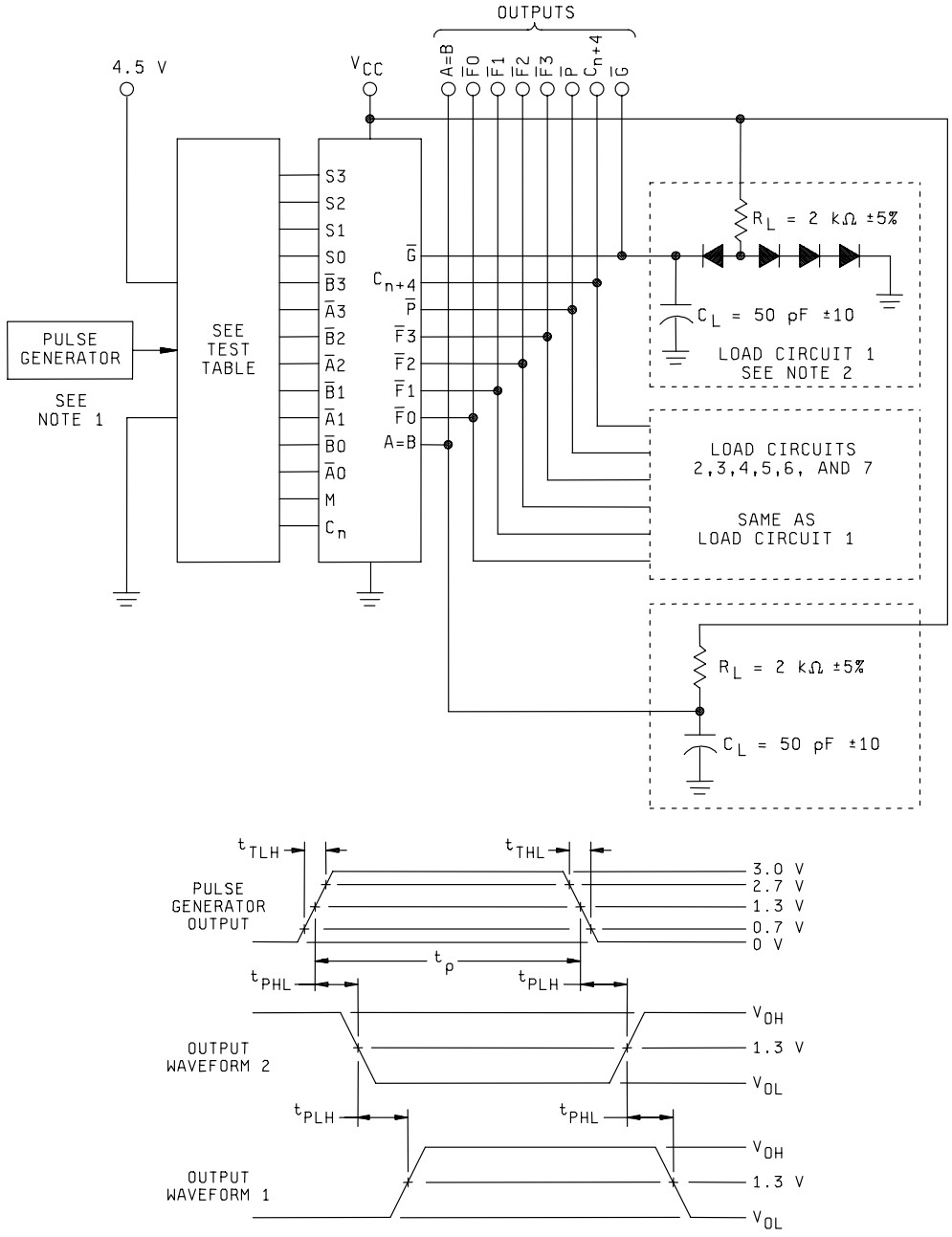


FIGURE 3. Logic diagram.

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NOTES:

1. The pulse generator has the following characteristics: PRR ≤ 1 MHz ±10%, Z_{OUT} ≈ 50 ohms, Pulse width = 200 ns ± 10%, t_{TLH} ≤ 15 ns, and t_{THL} ≤ 6 ns.
2. Load circuit on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.
3. C_L = includs probe and jig capacitance.
4. All diodes are 1N3064 or equivalent.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01.

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$\overline{SUM}$ MODE TEST TABLE

FUNCTION INPUTS: S0 = S3 = 4.5 V, S1 = S2 = M = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 4.5 V	APPLY GND	APPLY 4.5 V	APPLY GND		
t_{PLH1}	$\overline{A}$ i 1/	$\overline{B}$ i	None	Remaining $\overline{A}$ and $\overline{B}$	Cn	Any $\overline{F}$ i	1
t_{PHL1}							
t_{PLH1}	$\overline{B}$ i	$\overline{A}$ i	None	Remaining $\overline{A}$ and $\overline{B}$	Cn	Any $\overline{F}$ i	1
t_{PHL1}							
t_{PLH3}	$\overline{A}$ i	$\overline{B}$ i	None	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	$\overline{P}$	1
t_{PHL3}							
t_{PLH3}	$\overline{B}$ i	$\overline{A}$ i	None	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	$\overline{P}$	1
t_{PHL3}							
t_{PLH5}	$\overline{A}$ i	None	$\overline{B}$ i	Remaining $\overline{B}$	Remaining $\overline{A}$, Cn	$\overline{G}$	1
t_{PHL5}							
t_{PLH5}	$\overline{B}$ i	None	$\overline{A}$ i	Remaining $\overline{B}$	Remaining $\overline{A}$, Cn	$\overline{G}$	1
t_{PHL5}							
t_{PLH7}	Cn	None	None	All $\overline{A}$	All $\overline{B}$	Any $\overline{F}$ i	1
t_{PHL7}							
t_{PLH9}	Cn	None	None	All $\overline{A}$	All $\overline{B}$	Cn+4	1
t_{PHL9}							
t_{PLH11}	$\overline{A}$ i	None	$\overline{B}$ i	Remaining $\overline{B}$	Remaining $\overline{A}$, Cn	Cn+4	2
t_{PHL11}							
t_{PLH11}	$\overline{B}$ i	None	$\overline{A}$ i	Remaining $\overline{B}$	Remaining $\overline{A}$, Cn	Cn+4	2
t_{PHL11}							

1/ In $\overline{A}$ i, $\overline{B}$ i, and $\overline{F}$ i i = 0, 1, 2, 3.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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$\overline{\text{DIFF}}$ MODE TEST TABLE

FUNCTION INPUTS: S1 = S2 = 4.5 V, S0 = S3 = M = 0 V

TEST	INPUT UNDER TEST	OTHER INPUT SAME BIT		OTHER DATA INPUTS		OUTPUT UNDER TEST	OUTPUT WAVEFORM
		APPLY 4.5 V	APPLY GND	APPLY 4.5 V	APPLY GND		
t_{PLH2} / t_{PHL2}	$\overline{A}$ i 1/	None	$\overline{B}$ i	Remaining $\overline{A}$	Remaining $\overline{B}$, Cn	Any $\overline{F}$ i	1
t_{PLH2} / t_{PHL2}	$\overline{B}$ i	$\overline{A}$ i	None	Remaining $\overline{A}$	Remaining $\overline{B}$, Cn	Any $\overline{F}$ i	2
t_{PLH4} / t_{PHL4}	$\overline{A}$ i	None	$\overline{B}$ i	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	$\overline{P}$	1
t_{PLH4} / t_{PHL4}	$\overline{B}$ i	$\overline{A}$ i	None	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	$\overline{P}$	2
t_{PLH6} / t_{PHL6}	$\overline{A}$ i	$\overline{B}$ I	None	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	$\overline{G}$	1
t_{PLH6} / t_{PHL6}	$\overline{B}$ i	None	$\overline{A}$ i	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	$\overline{G}$	2
t_{PLH8} / t_{PHL8}	$\overline{A}$ i	None	$\overline{B}$ i	Remaining $\overline{A}$	Remaining $\overline{B}$, Cn	A = B	1
t_{PLH8} / t_{PHL8}	$\overline{B}$ I	$\overline{A}$ i	None	Remaining $\overline{A}$	Remaining $\overline{B}$, Cn	A = B	2
t_{PLH10} / t_{PHL10}	Cn	None	None	All $\overline{A}$ and $\overline{B}$	None	Cn+4	1
t_{PLH12} / t_{PHL12}	$\overline{A}$ i	$\overline{B}$ i	None	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	Cn+4	2
t_{PLH12} / t_{PHL12}	$\overline{B}$ i	None	$\overline{A}$ i	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	Cn+4	1
t_{PLH13} / t_{PHL13}	$\overline{A}$ i	None	$\overline{B}$ i	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	Any $\overline{F}$ i	1
t_{PLH13} / t_{PHL13}	$\overline{B}$ i	None	$\overline{A}$ i	None	Remaining $\overline{A}$ and $\overline{B}$, Cn	Any $\overline{F}$ i	2

1/ In $\overline{A}$ i, $\overline{B}$ i, and $\overline{F}$ i i = 0, 1, 2, 3.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01 - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
1	V_{OH1}	3006	1	0.7 V	2.0 V	2.0 V	0.7 V	0.7 V	0.7 V	2.0 V					"	$\overline{G}$	2.5		V
Tc = 25°C	V_{OH2}	"	2	2.0 V	"	"	2.0 V	0.7 V	0.7 V	0.7 V					"	Cn+4	"		"
	V_{OH3}	"	3	"	"	"	"	0.7 V	0.7 V	0.7 V					"	$\overline{P}$	"		"
	V_{OH4}	"	4	"	"	"	"	2.0 V	2.0 V	2.0 V	2.0 V	-400 µA			"	$\overline{F}$ 0	"		"
		"	5	"	"	"	"	"	"		0.7 V		-400 µA		"	$\overline{F}$ 1	"		"
		"	6	"	"	"	"	"	"		0.7 V			-400 µA	"	$\overline{F}$ 2	"		"
		"	7	"	"	"	"	"	"		0.7 V				"	$\overline{F}$ 3	"		"
	V_{OL1}	3007	8	"	"	"	"	0.7 V	"	0.7 V					"	$\overline{G}$		0.4	"
	V_{OL2}	"	9	"	"	"	0.7 V	0.7 V	0.7 V						"	Cn+4		"	"
	V_{OL3}	"	10	"	"	0.7 V	2.0 V	0.7 V	0.7 V	0.7 V					"	$\overline{P}$		"	"
	V_{OL4}	"	11	"	0.7 V	2.0 V	"	2.0 V	2.0 V	2.0 V	2.0 V	4 mA			"	$\overline{F}$ 0		"	"
		"	12	"	"	"	"	"	"	"	"		4 mA		"	$\overline{F}$ 1		"	"
		"	13	"	"	"	"	"	"	"	"			4 mA	"	$\overline{F}$ 2		"	"
		"	14	"	"	"	"	"	"	"	"				"	$\overline{F}$ 3		"	"
	V_{OL5}	"	15	"	"	"	"	"	"	"	"				"	A = B		"	"
	I_{IL1}	3009	16	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.4 V				"	M	5/	5/	mA
	I_{IL2}	"	17	0.4 V	"	"	"	"	"	"	5.5 V				"	$\overline{B}$ 0	"	"	"
		"	18	5.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 1	"	"	"
		"	19	"	"	"	"	"	"	"	"				"	$\overline{B}$ 2	"	"	"
		"	20	"	"	"	"	"	"	GND	"				"	$\overline{B}$ 3	"	"	"
	I_{IL3}	"	21	"	"	"	"	"	0.4 V	5.5 V	"				"	S0	"	"	"
		"	22	"	"	0.4 V	"	"	5.5 V	5.5 V	"				"	S3	"	"	"
	I_{IL4}	"	23	GND	"	GND	"	0.4 V	GND	GND	GND				"	S1	"	"	"
		"	24	"	"	"	0.4 V	GND	GND	GND	GND				"	S2	"	"	"
	I_{IL5}	"	25	"	GND	"	GND	GND	GND	0.4 V	GND				"	Cn	"	"	"
	I_{IL6}	"	26	"	0.4 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				"	$\overline{A}$ 0	"	"	"
		"	27	5.5 V	5.5 V	"	"	"	"	"	"				"	$\overline{A}$ 1	"	"	"
		"	28	5.5 V	5.5 V	"	"	"	"	"	"				"	$\overline{A}$ 2	"	"	"
		"	29	5.5 V	5.5 V	"	"	"	"	"	"				"	$\overline{A}$ 3	"	"	"
	I_{IH1}	3010	30	GND	GND	GND	GND	GND	GND	"	2.7 V				"	M		20	µA
	I_{IH2}	"	31	"	2.7 V	"	"	"	"	GND	GND				"	$\overline{A}$ 0		60	"
		"	32	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 1		"	"
		"	33	"	"	"	"	"	"	"	"				"	$\overline{A}$ 2		"	"
		"	34	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3		"	"

See footnotes at end of table. Pins 13 thru 24 on next page.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
1	V_{OH1}	3006	1					-400 µA	0.7 V	2.0 V	0.7 V	2.0 V	0.7 V	2.0 V	4.5 V	$\overline{G}$	2.5		V
Tc = 25°C	V_{OH2}	"	2				-400 µA		2.0 V	0.7 V	2.0 V	"	2.0 V	"	"	Cn+4	"		"
	V_{OH3}	"	3			-400 µA			"	2.0 V	2.0 V	"	"	"	"	$\overline{P}$	"		"
	V_{OH4}	"	4						"	0.7 V	2.0 V	"	"	"	"	$\overline{F}$ 0	"		"
		"	5						"	2.0 V	"	"	"	"	"	$\overline{F}$ 1	"		"
		"	6						"	2.0 V	"	"	"	"	"	$\overline{F}$ 2	"		"
		"	7	-400 µA					"	2.0 V	"	"	"	"	"	$\overline{F}$ 3	"		"
	V_{OL1}	3007	8					4 mA	0.7 V	0.7 V	2.0 V	2.0 V	"	"	"	$\overline{G}$		0.4	"
	V_{OL2}	"	9				4 mA		0.7 V	2.0 V	0.7 V	2.0 V	0.7 V	"	"	Cn+4		"	"
	V_{OL3}	"	10			4 mA			2.0 V	"	2.0 V	2.0 V	2.0 V	"	"	$\overline{P}$		"	"
	V_{OL4}	"	11						"	"	2.0 V	0.7 V	"	"	"	$\overline{F}$ 0		"	"
		"	12						"	"	"	0.7 V	"	0.7 V	"	$\overline{F}$ 1		"	"
		"	13						"	"	"	0.7 V	"	2.0 V	"	$\overline{F}$ 2		"	"
		"	14	4 mA					"	0.7 V	"	2.0 V	"	2.0 V	"	$\overline{F}$ 3		"	"
	V_{OL5}	"	15		4 mA				"	2.0 V	"	2.0 V	"	2.0 V	"	A = B		"	"
	I_{IL1}	3009	16						5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	M	5/	5/	mA
	I_{IL2}	"	17						"	"	5.5 V	"	5.5 V	"	"	$\overline{B}$ 0	"	"	"
		"	18						"	"	5.5 V	"	0.4 V	"	"	$\overline{B}$ 1	"	"	"
		"	19						"	"	0.4 V	"	5.5 V	"	"	$\overline{B}$ 2	"	"	"
		"	20						0.4 V	"	5.5 V	"	"	"	"	$\overline{B}$ 3	"	"	"
	I_{IL3}	"	21						5.5 V	"	5.5 V	"	"	"	"	S0	"	"	"
		"	22						5.5 V	"	5.5 V	"	"	"	"	S3	"	"	"
	I_{IL4}	"	23						GND	"	GND	"	GND	"	"	S1	"	"	"
		"	24						GND	"	GND	"	GND	"	"	S2	"	"	"
	I_{IL5}	"	25						GND	GND	GND	GND	GND	GND	"	Cn	"	"	"
	I_{IL6}	"	26						5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	$\overline{A}$ 0	"	"	"
		"	27						5.5 V	5.5 V	5.5 V	5.5 V	GND	0.4 V	"	$\overline{A}$ 1	"	"	"
		"	28						5.5 V	5.5 V	GND	0.4 V	5.5 V	5.5 V	"	$\overline{A}$ 2	"	"	"
		"	29						GND	0.4 V	5.5 V	5.5 V	5.5 V	5.5 V	"	$\overline{A}$ 3	"	"	"
	I_{IH1}	3010	30						"	GND	GND	GND	GND	GND	"	M		20	µA
	I_{IH2}	"	31						"	"	"	GND	"	GND	"	$\overline{A}$ 0		60	"
		"	32						"	"	"	GND	"	2.7 V	"	$\overline{A}$ 1		"	"
		"	33						"	"	"	2.7 V	"	GND	"	$\overline{A}$ 2		"	"
		"	34						"	2.7 V	"	GND	"	"	"	$\overline{A}$ 3		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
1	I_{IH2}	3010	35	2.7 V	GND	GND	GND	GND	GND	GND	GND				GND	$\overline{B}$ 0		60	µA
Tc = 25°C		"	36	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1		"	"
		"	37	"	"	"	"	"	"	"	"				"	$\overline{B}$ 2		"	"
		"	38	"	"	"	"	"	"	5.5 V	"				"	$\overline{B}$ 3		"	"
	I_{IH3}	"	39	"	"	"	"	"	2.7 V	GND	"				"	S0		80	"
		"	40	"	"	2.7 V	"	"	GND	"					"	S3		"	"
	I_{IH4}	"	41	5.5 V	"	GND	"	2.7 V	GND	"	"				"	S1		"	"
		"	42	5.5 V	"	GND	2.7 V	GND	GND	"	"				"	S2		"	"
	I_{IH5}	"	43	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	2.7 V	5.5 V				"	Cn		100	"
	I_{IH6}	"	44	GND	GND	GND	GND	GND	GND	GND	5.5 V				"	M		100	"
	I_{IH7}	"	45	"	5.5 V	"	"	"	"	"	GND				"	$\overline{A}$ 0		300	"
		"	46	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 1		"	"
		"	47	"	"	"	"	"	"	"	"				"	$\overline{A}$ 2		"	"
		"	48	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3		"	"
		"	49	5.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 0		"	"
		"	50	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1		"	"
		"	51	"	"	"	"	"	"	"	"				"	$\overline{B}$ 2		"	"
		"	52	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3		"	"
	I_{IH8}	"	53	"	"	"	"	"	5.5 V	"	"				"	S0		400	"
		"	54	"	"	5.5 V	"	"	GND	"	"				"	S3		"	"
	I_{IH9}	"	55	5.5 V	"	GND	"	5.5 V	GND	"	"				"	S1		"	"
		"	56	5.5 V	"	GND	5.5 V	GND	GND	"	"				"	S2		"	"
	I_{IH10}	"	57	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				"	Cn		500	"
	I_{OS1}	3011	58 ckts A, B & C	GND	GND	5.5 V	GND	GND	GND	GND	GND				"	$\overline{G}$	-15	-100	mA
	I_{OS2}	"	59 ckts A, B & C	GND	GND	GND	GND	GND	GND	GND	GND				"	Cn+4	"	"	"
	I_{OS3}	"	60 ckts A, B & C	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				"	$\overline{P}$	"	"	"
		"	61 ckts A, B & C	"	"	"	"	"	"	"	"	GND			"	$\overline{F}$ 0	"	"	"
		"	62 ckts A, B & C	"	"	"	"	"	"	"	"		GND		"	$\overline{F}$ 1	"	"	"
		"	63 ckts A, B & C	"	"	"	"	"	"	"	"			GND	"	$\overline{F}$ 2	"	"	"
		"	64 ckts A, B & C	"	"	"	"	"	"	"	"				"	$\overline{F}$ 3	"	"	"

See footnotes at end of table. Pins 13 thru 24 on next page.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
1	I_{IH2}	3010	35						GND	GND	GND	GND	GND	GND	5.5 V	$\overline{B}$ 0		60	µA
Tc = 25°C		"	36						"	"	GND	"	2.7 V	"	"	$\overline{B}$ 1		"	"
		"	37						"	"	2.7 V	"	GND	"	"	$\overline{B}$ 2		"	"
		"	38						2.7 V	"	GND	"	"	"	"	$\overline{B}$ 3		"	"
	I_{IH3}	"	39						GND	"	GND	"	"	"	"	S0		80	"
		"	40						GND	"	GND	"	"	"	"	S3		"	"
	I_{IH4}	"	41						5.5 V	"	5.5 V	"	5.5 V	"	"	S1		"	"
		"	42						5.5 V	"	5.5 V	"	5.5 V	"	"	S2		"	"
	I_{IH5}	"	43						5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	Cn		100	"
	I_{IH6}	"	44						GND	GND	GND	GND	GND	GND	"	M		100	"
	I_{IH7}	"	45						"	"	"	"	"	GND	"	$\overline{A}$ 0		300	"
		"	46						"	"	"	"	"	5.5 V	"	$\overline{A}$ 1		"	"
		"	47						"	"	"	5.5 V	"	GND	"	$\overline{A}$ 2		"	"
		"	48						"	5.5 V	"	GND	"	"	"	$\overline{A}$ 3		"	"
		"	49						"	GND	"	"	"	"	"	$\overline{B}$ 0		"	"
		"	50						"	"	"	"	5.5 V	"	"	$\overline{B}$ 1		"	"
		"	51						"	"	5.5 V	"	GND	"	"	$\overline{B}$ 2		"	"
		"	52						5.5 V	"	GND	"	"	"	"	$\overline{B}$ 3		"	"
	I_{IH8}	"	53						GND	"	"	"	"	"	"	S0		400	"
		"	54						GND	"	"	"	"	"	"	S3		"	"
	I_{IH9}	"	55						5.5 V	"	5.5 V	"	5.5 V	"	"	S1		"	"
		"	56						5.5 V	"	5.5 V	"	5.5 V	"	"	S2		"	"
	I_{IH10}	"	57						5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	Cn		500	"
	I_{OS1}	3011	58 ckts A, B & C					GND	5.5 V	5.5 V	GND	GND	GND	GND	"	$\overline{G}$	-15	-100	mA
	I_{OS2}	"	59 ckts A, B & C				GND		GND	GND	GND	GND	GND	GND	"	Cn+4	"	"	"
	I_{OS3}	"	60 ckts A, B & C			GND			5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	$\overline{P}$	"	"	"
		"	61 ckts A, B & C						"	"	"	"	"	"	"	$\overline{F}$ 0	"	"	"
		"	62 ckts A, B & C						"	"	"	"	"	"	"	$\overline{F}$ 1	"	"	"
		"	63 ckts A, B & C						"	"	"	"	"	"	"	$\overline{F}$ 2	"	"	"
		"	64 ckts A, B & C	GND					"	"	"	"	"	"	"	$\overline{F}$ 3	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
1	I_{CCL1}	3005	65	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V				GND	V_{CC}		32	mA
Tc = 25°C	I_{CCH1}	"	66	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V				"	V_{CC}		35	mA
	V_{IC}	"	67	-18 mA											"	$\overline{B}$ 0		-1.5	V
		"	68		-18 mA										"	$\overline{A}$ 0		"	"
		"	69			-18 mA									"	S3		"	"
		"	70				-18 mA								"	S2		"	"
		"	71					-18 mA							"	S1		"	"
		"	72						-18 mA						"	S0		"	"
		"	73							-18 mA					"	Cn		"	"
		"	74								-18 mA				"	M		"	"
		"	75												"	$\overline{B}$ 3		"	"
		"	76												"	$\overline{A}$ 3		"	"
		"	77												"	$\overline{B}$ 2		"	"
		"	78												"	$\overline{A}$ 2		"	"
		"	79												"	$\overline{B}$ 1		"	"
		"	80												"	$\overline{A}$ 1		"	"
	I_{CEX}	"	81		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V		2.0 V				"	A = B		100	µA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C, and V_{IC} tests are omitted.																		
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																		
7 1/ 2/ 4/	Func-	3014	82	B	B	B	B	B	B	B	B	H 3/	L 3/	L 3/	GND	All	H or L		
Tc = 25°C	tional	"	83	B	B	B	A	"	A	A	A	"	H	H	"	outputs	as shown 3/		
	tests	"	84	B	B	A	A	"	B	A	A	"	H	H	"	"			
	"	"	85	A	A	A	B	"	A	B	B	"	L	L	"	"			
	"	"	86	B	A	B	A	"	"	B	"	"	H	"	"	"			
	"	"	87	B	B	A	A	A	"	B	"	L	L	"	"	"			
	"	"	88	A	"	"	A	B	"	A	"	H	H	"	"	"			
	"	"	89	B	"	"	B	B	B	A	A	H	H	H	"	"			
	"	"	90	"	"	"	A	B	"	A	B	L	L	"	"	"			
	"	"	91	"	A	A	A	A	"	B	B	H	H	"	"	"			
	"	"	92	"	B	B	A	B	"	A	A	H	H	"	"	"			
	"	"	93	"	B	A	B	"	A	B	B	H	H	L	"	"			
	"	"	94	"	A	A	A	"	A	A	"	L	L	H	"	"			
	"	"	95	"	B	A	"	"	B	A	"	L	L	"	"	"			
	"	"	96	"	A	B	"	"	A	B	"	H	H	"	"	"			
	"	"	97	"	B	B	"	"	A	A	"	L	H	"	"	"			
	"	"	98	"	A	A	"	"	A	A	"	L	L	L	"	"			
	"	"	99	"	A	B	"	A	B	B	"	H	H	H	"	"			
	"	"	100	"	A	A	"	A	A	B	"	H	L	L	"	"			
	"	"	101	"	B	B	"	B	B	A	"	L	"	L	"	"			
	"	"	102	"	B	A	"	B	A	"	"	"	"	L	"	"			
	"	"	103	"	B	"	B	B	"	"	"	"	"	H	"	"			
	"	"	104	"	A	"	A	A	"	"	"	"	"	L	"	"			
	"	"	105	A	B	"	B	B	"	"	"	H	H	H	"	"			
	"	"	106	B	B	"	A	B	"	"	"	L	L	L	"	"			
	"	"	107	A	A	B	A	B	B	"	"	H	H	H	"	"			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
1	I_{CCL1}	3005	65						GND	5.5 V	GND	5.5 V	GND	5.5 V	5.5 V	V_{CC}		32	mA
Tc = 25°C	I_{CCH1}	"	66						GND	GND	GND	GND	GND	GND	5.5 V	V_{CC}		35	mA
	V_{IC}	"	67												4.5 V	$\overline{B}$ 0		-1.5	V
		"	68												"	$\overline{A}$ 0		"	"
		"	69												"	S3		"	"
		"	70												"	S2		"	"
		"	71												"	S1		"	"
		"	72												"	S0		"	"
		"	73												"	Cn		"	"
		"	74												"	M		"	"
		"	75						-18 mA						"	$\overline{B}$ 3		"	"
		"	76							-18 mA					"	$\overline{A}$ 3		"	"
		"	77								-18 mA				"	$\overline{B}$ 2		"	"
		"	78									-18 mA			"	$\overline{A}$ 2		"	"
		"	79										-18 mA		"	$\overline{B}$ 1		"	"
		"	80											-18 mA	"	$\overline{A}$ 1		"	"
	I_{CEX}	"	81		5.5 V					2.0 V		2.0 V		2.0 V	"	A = B		100	µA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C, and V_{IC} tests are omitted.																		
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																		
7 1/ 2/ 4/	Func-	3014	82	L 3/	L 3/	L 3/	H 3/	L 3/	B	B	B	B	B	B	4.5 V	All	H or L		
Tc = 25°C	tional	"	83	H	H	L	H	L	B	B	B	"	"	B	"	outputs	as shown 3/		
	tests	"	84	H	H	H	L	H	B	A	B	"	"	B	"	"			
	"	"	85	L	L	H	"	"	A	B	A	"	"	A	"	"			
	"	"	86	"	"	H	"	"	A	"	B	A	"	B	"	"			
	"	"	87	"	"	L	"	"	A	"	A	B	A	"	"	"			
	"	"	88	"	"	L	H	L	B	"	B	B	A	"	"	"			
	"	"	89	"	"	L	"	"	"	A	B	B	B	"	"	"			
	"	"	90	H	"	H	"	"	"	B	B	A	"	A	"	"			
	"	"	91	H	H	H	"	"	"	B	A	B	"	A	"	"			
	"	"	92	L	L	L	"	"	A	A	B	"	"	B	"	"			
	"	"	93	H	"	L	"	"	A	B	"	"	A	B	"	"			
	"	"	94	L	"	H	"	"	B	"	"	"	A	B	"	"			
	"	"	95	L	"	H	"	"	A	"	"	"	B	A	"	"			
	"	"	96	H	H	H	"	"	A	"	A	"	B	B	"	"			
	"	"	97	"	L	L	"	"	A	A	A	A	A	A	"	"			
	"	"	98	"	L	H	"	"	B	B	A	B	A	B	"	"			
	"	"	99	"	H	"	"	"	B	"	B	B	A	"	"	"			
	"	"	100	L	L	"	L	H	A	"	A	B	B	"	"	"			
	"	"	101	H	"	"	H	L	B	"	B	A	B	"	"	"			
	"	"	102	H	"	"	H	L	B	"	A	B	A	A	"	"			
	"	"	103	H	"	L	H	L	A	"	B	A	B	B	"	"			
	"	"	104	L	"	H	L	H	"	"	A	B	A	"	"	"			
	"	"	105	H	H	L	H	"	"	"	"	B	A	"	"	"			
	"	"	106	L	L	H	L	"	"	"	"	A	B	"	"	"			
	"	"	107	H	H	L	H	"	"	A	"	A	A	A	"	"			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
7 1/ 2/ 4/ Tc = 25°C	Func-tional tests	3014	108	B	B	A	B	B	B	A	A	H	H	H	GND	All	H or L as shown 3/		
	"	"	109	B	A	B	B	B	A	A	B	H	H	H	"	outputs			
	"	"	110	A	B	A	B	B	B	B	A	H	H	H	"	"			
	"	"	111	B	A	A	B	A	A	A	A	L	H	L	"	"			
	"	"	112	B	A	A	A	A	B	B	A	H	L	L	"	"			
	"	"	113	A	B	A	A	A	B	A	B	L	L	L	"	"			
	"	"	114	B	B	B	B	B	B	B	B	H	H	H	"	"			
	"	"	115	"	A	A	A	A	A	B	A	H	L	H	"	"			
	"	"	116	"	B	A	B	B	A	B	B				"	"			
	"	"	117	"	A							L	L	L	"	"			
	"	"	118	"	B							H	H	H	"	"			
	"	"	119	A	B							L	L	L	"	"			
	"	"	120	B	B							H	H	H	"	"			
8 1/ 2/ 4/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = -55°C																		
9	t_{PHL1}	3003	121	GND	IN	4.5 V	GND	GND	4.5 V	GND	GND	OUT			GND	$\overline{A}$ 0 to $\overline{F}$ 0	2	25	ns
Tc = 25°C		Fig 4	122	"	GND	"	"	"	"	"	"		OUT		"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	123	"	"	"	"	"	"	"	"			OUT	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	124	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	125	"	"	"	"	"	"	"	"	OUT			"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	126	"	"	"	"	"	"	"	"		OUT		"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	127	"	"	"	"	"	"	"	"			OUT	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	128	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PLH1}	"	129	"	IN	"	"	"	"	"	"	OUT			"	$\overline{A}$ 0 to $\overline{F}$ 0	"	55	"
		"	130	"	GND	"	"	"	"	"	"		OUT		"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	131	"	"	"	"	"	"	"	"			OUT	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	132	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	133	"	"	"	"	"	"	"	"	OUT			"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	134	"	"	"	"	"	"	"	"		OUT		"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	135	"	"	"	"	"	"	"	"			OUT	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	136	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PHL2}	"	137	"	IN	GND	4.5 V	4.5 V	GND	"	"	OUT			"	$\overline{A}$ 0 to $\overline{F}$ 0	"	30	"
		"	138	"	4.5 V	"	"	"	"	"	"		OUT		"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	139	"	"	"	"	"	"	"	"			OUT	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	140	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	141	IN	"	"	"	"	"	"	"	OUT			"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	142	GND	"	"	"	"	"	"	"		OUT		"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	143	"	"	"	"	"	"	"	"			OUT	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	144	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
7 1/ 2/ 4/ Tc = 25°C	Func-tional tests	3014	108	H	H	H	L	H	A	A	B	B	B	B	4.5 V	All	H or L as shown 3/		
	"	"	109	"	H	L	H	"	"	B	A	B	A	B	"	outputs			
	"	"	110	"	H	H	L	"	"	A	A	A	B	B	"	"			
	"	"	111	"	L	"	L	"	"	A	A	B	A	A	"	"			
	"	"	112	"	"	"	H	L	"	B	B	B	B	B	"	"			
	"	"	113	L	"	"	L	H	B	B	"	B	A	A	"	"			
	"	"	114	L	"	L	H	L	"	B	"	A	B	A	"	"			
	"	"	115	H	"	H	L	H	"	A	"	A	"	B	"	"			
	"	"	116						"	B	"	B	"	B	"	"			
	"	"	117	L					"	A	"	A	"	A	"	"			
	"	"	118	H					"	B	"	B	"	B	"	"			
	"	"	119	L					A	B	A	B	A	B	"	"			
	"	"	120	H					B	B	B	B	B	B	"	"			
8 1/ 2/ 4/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = -55°C																		
9	t_{PHL1}	3003	121						GND	GND	GND	GND	GND	GND	5.0 V	$\overline{A}$ 0 to $\overline{F}$ 0	2	25	ns
Tc = 25°C		Fig 4	122						"	GND	"	GND	"	IN	"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	123						"	GND	"	IN	"	GND	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	124	OUT					"	IN	"	GND	"	"	"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	125						"	GND	"	"	"	"	"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	126						"	"	"	"	IN	"	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	127						"	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	128	OUT					IN	"	GND	"	"	"	"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PLH1}	"	129						GND	"	"	"	"	"	"	$\overline{A}$ 0 to $\overline{F}$ 0	"	55	"
		"	130						"	"	"	"	"	IN	"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	131						"	"	"	IN	"	GND	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	132	OUT					"	IN	"	GND	"	"	"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	133						"	GND	"	"	"	"	"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	134						"	"	"	'	IN	"	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	135						"	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	136	OUT					IN	"	GND	"	"	"	"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PHL2}	"	137						GND	4.5 V	GND	4.5 V	"	4.5 V	"	$\overline{A}$ 0 to $\overline{F}$ 0	"	30	"
		"	138						"	4.5 V	"	4.5 V	"	IN	"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	139						"	4.5 V	"	IN	"	4.5 V	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	140	OUT					"	IN	"	4.5 V	"	"	"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	141						"	4.5 V	"	"	"	"	"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	142						"	"	"	"	IN	"	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	143						"	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	144	OUT					IN	"	GND	"	"	"	"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
9	t_{PLH2}	3003	145	GND	IN	GND	4.5 V	4.5 V	GND	GND	GND	OUT			GND	$\overline{A}$ 0 to $\overline{F}$ 0	2	32	ns
Tc = 25°C		Fig 4	146	"	4.5 V	"	"	"	"	"	"		OUT		"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	147	"	"	"	"	"	"	"	"			OUT	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	148	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	149	IN	"	"	"	"	"	"	"	OUT			"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	150	GND	"	"	"	"	"	"	"		OUT		"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	151	GND	"	"	"	"	"	"	"			OUT	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	152	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PHL3}	"	153	4.5 V	IN	4.5 V	GND	GND	4.5 V	"	"				"	$\overline{A}$ 0 to $\overline{P}$	"	"	"
		"	154	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	155	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	156	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	157	IN	4.5 V	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	158	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	159	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	160	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{P}$	"	"	"
	t_{PLH3}	"	161	4.5 V	IN	"	"	"	"	"	"				"	$\overline{A}$ 0 to $\overline{P}$	"	30	"
		"	162	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	163	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	164	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	165	IN	4.5 V	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	166	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	167	"	GND	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	168	"	GND	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{P}$	"	"	"
	t_{PHL4}	"	169	"	IN	GND	4.5 V	4.5 V	GND	GND	GND				"	$\overline{A}$ 0 to $\overline{P}$	"	33	"
		"	170	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	171	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	172	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	173	IN	4.5 V	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	174	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	175	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	176	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{P}$	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
9	t_{PLH2}	3003	145						GND	GND	GND	GND	GND	GND	5.0 V	$\overline{A}$ 0 to $\overline{F}$ 0	2	32	ns
Tc = 25°C		Fig 4	146						"	GND	"	GND	"	IN	"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	147						"	GND	"	IN	"	4.5 v	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	148	OUT					"	IN	"	4.5 v	"	"	"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	149						"	4.5 v	"	"	"	"	"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	150						"	"	"	"	IN	"	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	151						"	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	152	OUT					IN	"	GND	"	GND	"	"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PHL3}	"	153			OUT			GND	GND	GND	GND	GND	GND	"	$\overline{A}$ 0 to $\overline{P}$	"	"	"
		"	154			"			GND	GND	GND	GND	4.5 V	IN	"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	155			"			GND	GND	4.5 V	IN	GND	GND	"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	156			"			4.5 V	IN	GND	GND	GND	GND	"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	157			"			GND	GND	GND	GND	GND	GND	"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	158			"			GND	GND	GND	GND	IN	4.5 V	"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	159			"			GND	GND	IN	4.5 V	GND	GND	"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	160			"			IN	4.5 V	GND	GND	GND	GND	"	$\overline{B}$ 3 to $\overline{P}$	"	"	"
	t_{PLH3}	"	161			"			GND	GND	GND	GND	GND	GND	"	$\overline{A}$ 0 to $\overline{P}$	"	30	"
		"	162			"			GND	GND	GND	GND	4.5 V	IN	"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	163			"			GND	GND	4.5 V	IN	GND	GND	"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	164			"			4.5 V	IN	GND	GND	GND	GND	"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	165			"			GND	GND	GND	GND	GND	GND	"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	166			"			GND	GND	GND	GND	IN	4.5 V	"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	167			"			GND	GND	IN	4.5 V	GND	GND	"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	168			"			IN	4.5 V	GND	GND	"	GND	"	$\overline{B}$ 3 to $\overline{P}$	"	"	"
	t_{PHL4}	"	169			"			GND	GND	"	GND	"	GND	"	$\overline{A}$ 0 to $\overline{P}$	"	33	"
		"	170			"			"	GND	"	GND	"	IN	"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	171			"			"	GND	"	IN	"	GND	"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	172			"			"	IN	"	GND	"	GND	"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	173			"			"	GND	"	GND	"	GND	"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	174			"			"	GND	"	GND	IN	4.5 V	"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	175			"			"	GND	IN	4.5 V	GND	GND	"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	176			"			IN	4.5 V	GND	GND	GND	GND	"	$\overline{B}$ 3 to $\overline{P}$	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
9	t_{PLH4}	3003	177	GND	IN	GND	4.5 V	4.5 V	GND	GND	GND				GND	$\overline{A}$ 0 to $\overline{P}$	2	30	NS
Tc = 25°C		Fig 4	178	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	179	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	180	"	GND	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	181	IN	4.5 V	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	182	GND	GND	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	183	"	GND	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	184	"	GND	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{P}$	"	"	"
	t_{PHL5}	"	185	"	IN	4.5 V	GND	GND	4.5 V	"	"				"	$\overline{A}$ 0 to $\overline{G}$	"	28	"
		"	186	4.5 V	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	187	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	188	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	189	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	190	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	191	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	192	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{G}$	"	"	"
	t_{PLH5}	"	193	GND	IN	"	"	"	"	"	"				"	$\overline{A}$ 0 to $\overline{G}$	"	32	"
		"	194	4.5 V	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	195	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	196	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	197	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	198	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	199	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	200	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{G}$	"	"	"
	t_{PHL6}	"	201	4.5 V	IN	GND	4.5 V	4.5 V	GND	"	"				"	$\overline{A}$ 0 to $\overline{G}$	"	30	"
		"	202	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	203	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	204	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	205	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	206	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	207	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	208	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{G}$	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
9	t_{PLH4}	3003	177			OUT			GND	GND	GND	GND	GND	GND	5.0 V	$\overline{A}$ 0 to $\overline{P}$	2	30	ns
Tc = 25°C		Fig 4	178			"			"	GND	"	GND	"	IN	"	$\overline{A}$ 1 to $\overline{P}$	"	"	"
		"	179			"			"	GND	"	IN	"	GND	"	$\overline{A}$ 2 to $\overline{P}$	"	"	"
		"	180			"			"	IN	"	GND	"	GND	"	$\overline{A}$ 3 to $\overline{P}$	"	"	"
		"	181			"			"	GND	"	GND	"	GND	"	$\overline{B}$ 0 to $\overline{P}$	"	"	"
		"	182			"			"	GND	"	GND	IN	4.5 V	"	$\overline{B}$ 1 to $\overline{P}$	"	"	"
		"	183			"			"	GND	IN	4.5 V	GND	GND	"	$\overline{B}$ 2 to $\overline{P}$	"	"	"
		"	184			"			IN	4.5 V	GND	GND	GND	GND	"	$\overline{B}$ 3 to $\overline{P}$	"	"	"
	t_{PHL5}	"	185					OUT	4.5 V	GND	4.5 V	GND	4.5 V	GND	"	$\overline{A}$ 0 to $\overline{G}$	"	28	"
		"	186					"	4.5 V	GND	4.5 V	GND	GND	IN	"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	187					"	4.5 V	GND	GND	IN	4.5 V	GND	"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	188					"	GND	IN	4.5 V	GND	4.5 V	"	"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	189					"	4.5 V	GND	4.5 V	"	4.5 V	"	"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	190					"	4.5 V	"	4.5 V	"	IN	"	"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	191					"	4.5 V	"	IN	"	4.5 V	"	"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	192					"	IN	"	4.5 V	"	4.5 V	"	"	$\overline{B}$ 3 to $\overline{G}$	"	"	"
	t_{PLH5}	"	193					"	4.5 V	"	4.5 V	"	4.5 V	"	"	$\overline{A}$ 0 to $\overline{G}$	"	32	"
		"	194					"	4.5 V	"	4.5 V	"	GND	IN	"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	195					"	4.5 V	"	GND	IN	4.5 V	GND	"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	196					"	GND	IN	4.5 V	GND	4.5 V	"	"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	197					"	4.5 V	GND	4.5 V	"	4.5 V	"	"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	198					"	4.5 V	"	4.5 V	"	IN	"	"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	199					"	4.5 V	"	IN	"	4.5 V	"	"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	200					"	IN	"	4.5 V	"	4.5 V	"	"	$\overline{B}$ 3 to $\overline{G}$	"	"	"
	t_{PHL6}	"	201					"	GND	"	GND	"	GND	"	"	$\overline{A}$ 0 to $\overline{G}$	"	30	"
		"	202					"	GND	"	GND	"	4.5 V	IN	"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	203					"	GND	"	4.5 V	IN	GND	GND	"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	204					"	4.5 V	IN	GND	GND	GND	"	"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	205					"	GND	GND	GND	"	GND	"	"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	206					"	GND	"	GND	"	IN	"	"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	207					"	GND	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	208					"	IN	"	GND	"	GND	"	"	$\overline{B}$ 3 to $\overline{G}$	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
9	t_{PLH6}	3003	209	4.5 V	IN	GND	4.5 V	4.5 V	GND	GND	GND				GND	$\overline{A}$ 0 to $\overline{G}$	2	32	ns
Tc = 25°C		Fig 4	210	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	211	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	212	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	213	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	214	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	215	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	216	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{G}$	"	"	"
	t_{PHL7}	"	217	"	4.5 V	4.5 V	GND	GND	4.5 V	IN	"	OUT			"	Cn to $\overline{F}$ 0	"	25	"
	t_{PLH7}	"	218	"	4.5 V	4.5 V	GND	GND	4.5 V	IN	"	OUT			"	Cn to $\overline{F}$ 0	"	26	"
	t_{PHL8}	"	219	"	IN	GND	4.5 V	4.5 V	GND	GND	"				"	$\overline{A}$ 0 to A = B	"	75	"
		"	220	"	4.5 V	"	"	"	"	"	"				"	$\overline{A}$ 1 to A = B	"	"	"
		"	221	"	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to A = B	"	"	"
		"	222	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to A = B	"	"	"
		"	223	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to A = B	"	"	"
		"	224	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to A = B	"	"	"
		"	225	"	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to A = B	"	"	"
		"	226	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to A = B	"	"	"
	t_{PHL8}	"	227	"	IN	"	"	"	"	"	"				"	$\overline{A}$ 0 to A = B	"	80	"
		"	228	"	4.5 V	"	"	"	"	"	"				"	$\overline{A}$ 1 to A = B	"	"	"
		"	229	"	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to A = B	"	"	"
		"	230	"	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to A = B	"	"	"
		"	231	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to A = B	"	"	"
		"	232	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to A = B	"	"	"
		"	233	"	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to A = B	"	"	"
		"	234	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to A = B	"	"	"
	t_{PHL9}	"	235	"	"	4.5 V	GND	GND	4.5 V	IN	"				"	Cn to Cn+4	"	25	"
	t_{PLH9}	"	236	"	"	4.5 V	GND	GND	4.5 V	"	"				"	Cn to Cn+4	"	26	"
	t_{PHL10}	"	237	4.5 V	"	GND	4.5 V	4.5 V	GND	"	"				"	Cn to Cn+4	"	25	"
	t_{PLH10}	"	238	4.5 V	"	GND	4.5 V	4.5 V	GND	"	"				"	Cn to Cn+4	"	26	"

See footnotes at end of table. Pins 13 thru 24 on next page.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
9	t_{PLH6}	3003	209					OUT	GND	GND	GND	GND	GND	GND	5.0 V	$\overline{A}$ 0 to $\overline{G}$	2	32	ns
Tc = 25°C		Fig 4	210					"	GND	GND	"	"	4.5 V	IN	"	$\overline{A}$ 1 to $\overline{G}$	"	"	"
		"	211					"	GND	GND	4.5 V	IN	GND	GND	"	$\overline{A}$ 2 to $\overline{G}$	"	"	"
		"	212					"	4.5 V	IN	GND	GND	GND	"	"	$\overline{A}$ 3 to $\overline{G}$	"	"	"
		"	213					"	GND	GND	GND	"	GND	"	"	$\overline{B}$ 0 to $\overline{G}$	"	"	"
		"	214					"	GND	"	GND	"	IN	"	"	$\overline{B}$ 1 to $\overline{G}$	"	"	"
		"	215					"	GND	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{G}$	"	"	"
		"	216					"	IN	"	GND	"	"	"	"	$\overline{B}$ 3 to $\overline{G}$	"	"	"
	t_{PHL7}	"	217						GND	4.5 V	"	4.5 V	"	4.5 V	"	Cn to $\overline{F}$ 0	"	25	"
	t_{PLH7}	"	218						"	"	"	"	"	4.5 V	"	Cn to $\overline{F}$ 0	"	26	"
	t_{PHL8}	"	219		OUT				"	"	"	"	"	4.5 V	"	$\overline{A}$ 0 to A = B	"	75	"
		"	220		"				"	"	"	"	"	IN	"	$\overline{A}$ 1 to A = B	"	"	"
		"	221		"				"	"	"	IN	"	4.5 V	"	$\overline{A}$ 2 to A = B	"	"	"
		"	222		"				"	IN	"	4.5 V	"	"	"	$\overline{A}$ 3 to A = B	"	"	"
		"	223		"				"	4.5 V	"	"	"	"	"	$\overline{B}$ 0 to A = B	"	"	"
		"	224		"				"	"	"	"	IN	"	"	$\overline{B}$ 1 to A = B	"	"	"
		"	225		"				"	"	IN	"	GND	"	"	$\overline{B}$ 2 to A = B	"	"	"
		"	226		"				IN	"	GND	"	"	"	"	$\overline{B}$ 3 to A = B	"	"	"
	t_{PHL8}	"	227		"				GND	"	"	"	"	"	"	$\overline{A}$ 0 to A = B	"	80	"
		"	228		"				"	"	"	"	"	IN	"	$\overline{A}$ 1 to A = B	"	"	"
		"	229		"				"	"	"	IN	"	4.5 V	"	$\overline{A}$ 2 to A = B	"	"	"
		"	230		"				"	IN	"	4.5 V	"	"	"	$\overline{A}$ 3 to A = B	"	"	"
		"	231		"				"	4.5 V	"	"	IN	"	"	$\overline{B}$ 0 to A = B	"	"	"
		"	232		"				"	"	IN	"	GND	"	"	$\overline{B}$ 1 to A = B	"	"	"
		"	233		"				"	"	GND	"	"	"	"	$\overline{B}$ 2 to A = B	"	"	"
		"	234		"				IN	"	"	"	"	"	"	$\overline{B}$ 3 to A = B	"	"	"
	t_{PHL9}	"	235				OUT		GND	"	"	"	"	"	"	Cn to Cn+4	"	25	"
	t_{PLH9}	"	236				"		GND	"	"	"	"	"	"	Cn to Cn+4	"	26	"
	t_{PHL10}	"	237				"		4.5 V	"	4.5 V	"	4.5 V	"	"	Cn to Cn+4	"	25	"
	t_{PLH10}	"	238				"		4.5 V	"	4.5 V	"	4.5 V	"	"	Cn to Cn+4	"	26	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	1	2	3	4	5	6	7	8	9	10	11	12	Measured terminal	Limits		Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
			Test no.	$\overline{B}$ 0	$\overline{A}$ 0	S3	S2	S1	S0	Cn	M	$\overline{F}$ 0	$\overline{F}$ 1	$\overline{F}$ 2	GND		Min	Max	
9	t_{PHL11}	3003	239	GND	IN	4.5 V	GND	GND	4.5 V	GND	GND				GND	$\overline{A}$ 0 to Cn+4	2	40	ns
Tc = 25°C		Fig 4	240	4.5 V	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	241	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	242	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	243	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	244	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	245	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	246	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to Cn+4	"	"	"
	t_{PLH11}	"	247	GND	IN	"	"	"	"	"	"				"	$\overline{A}$ 0 to Cn+4	"	38	"
		"	248	4.5 V	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	249	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	250	4.5 V	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	251	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	252	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	253	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	254	4.5 V	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to Cn+4	"	"	"
	t_{PHL12}	"	255	4.5 V	IN	GND	4.5 V	4.5 V	GND	"	"				"	$\overline{A}$ 0 to Cn+4	"	46	"
		"	256	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	257	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	258	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	259	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	260	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	261	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	262	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to Cn+4	"	"	"
	t_{PLH12}	"	263	4.5 V	IN	"	"	"	"	"	"				"	$\overline{A}$ 0 to Cn+4	"	40	"
		"	264	GND	GND	"	"	"	"	"	"				"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	265	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	266	GND	"	"	"	"	"	"	"				"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	267	IN	"	"	"	"	"	"	"				"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	268	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	269	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	270	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to Cn+4	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
9	t_{PHL11}	3003	239				OUT		4.5 V	GND	4.5 V	GND	4.5 V	GND	5.0 V	$\overline{A}$ 0 to Cn+4	2	40	ns
Tc = 25°C		Fig 4	240				"		"	GND	"	GND	GND	IN	"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	241				"		"	GND	GND	IN	4.5 V	GND	"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	242				"		GND	IN	4.5 V	GND	4.5 V	"	"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	243				"		4.5 V	GND	4.5 V	"	4.5 V	"	"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	244				"		4.5 V	"	4.5 V	"	IN	"	"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	245				"		4.5 V	"	IN	"	4.5 V	"	"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	246				"		IN	"	4.5 V	"	4.5 V	"	"	$\overline{B}$ 3 to Cn+4	"	"	"
	t_{PLH11}	"	247				"		4.5 V	"	4.5 V	"	4.5 V	"	"	$\overline{A}$ 0 to Cn+4	"	38	"
		"	248				"		4.5 V	"	4.5 V	"	GND	IN	"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	249				"		4.5 V	"	GND	IN	4.5 V	GND	"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	250				"		GND	IN	4.5 V	GND	4.5 V	"	"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	251				"		4.5 V	GND	4.5 V	"	4.5 V	"	"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	252				"		4.5 V	"	4.5 V	"	IN	"	"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	253				"		4.5 V	"	IN	"	4.5 V	"	"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	254				"		IN	"	4.5 V	"	4.5 V	"	"	$\overline{B}$ 3 to Cn+4	"	"	"
	t_{PHL12}	"	255				"		GND	"	GND	"	GND	"	"	$\overline{A}$ 0 to Cn+4	"	46	"
		"	256				"		GND	"	GND	"	4.5 V	IN	"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	257				"		GND	"	4.5 V	IN	GND	GND	"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	258				"		4.5 V	IN	GND	GND	GND	"	"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	259				"		GND	GND	GND	"	GND	"	"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	260				"		GND	"	GND	"	IN	"	"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	261				"		GND	"	IN	"	GND	"	"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	262				"		IN	"	GND	"	GND	"	"	$\overline{B}$ 3 to Cn+4	"	"	"
	t_{PLH12}	"	263				"		GND	"	GND	"	GND	"	"	$\overline{A}$ 0 to Cn+4	"	40	"
		"	264				"		GND	"	GND	"	4.5 V	IN	"	$\overline{A}$ 1 to Cn+4	"	"	"
		"	265				"		GND	"	4.5 V	IN	GND	GND	"	$\overline{A}$ 2 to Cn+4	"	"	"
		"	266				"		4.5 V	IN	GND	GND	GND	"	"	$\overline{A}$ 3 to Cn+4	"	"	"
		"	267				"		GND	GND	GND	"	GND	"	"	$\overline{B}$ 0 to Cn+4	"	"	"
		"	268				"		GND	"	GND	"	IN	"	"	$\overline{B}$ 1 to Cn+4	"	"	"
		"	269				"		GND	"	IN	"	GND	"	"	$\overline{B}$ 2 to Cn+4	"	"	"
		"	270				"		IN	"	GND	"	GND	"	"	$\overline{B}$ 3 to Cn+4	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z / Case 3 1/ / Test no.	1 / $\overline{B}$ 0	2 / $\overline{A}$ 0	3 / S3	4 / S2	5 / S1	6 / S0	7 / Cn	8 / M	9 / $\overline{F}$ 0	10 / $\overline{F}$ 1	11 / $\overline{F}$ 2	12 / GND	Measured terminal	Limits Min	Limits Max	Unit
			Case 3 1/	2	3	4	5	6	7	9	10	11	12	13	14				
9	t_{PHL13}	3003	271	GND	IN	GND	4.5 V	4.5 V	GND	GND	4.5 V	OUT			GND	$\overline{A}$ 0 to $\overline{F}$ 0	2	38	ns
Tc = 25°C		Fig 4	272	"	GND	"	"	"	"	"	"		OUT		"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	273	"	"	"	"	"	"	"	"			OUT	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	274	"	"	"	"	"	"	"	"	OUT			"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	275	IN	"	"	"	"	"	"	"		OUT		"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	276	GND	"	"	"	"	"	"	"			OUT	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	277	"	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	278	"	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PLH13}	"	279	"	IN	"	"	"	"	"	"	OUT			"	$\overline{A}$ 0 to $\overline{F}$ 0	"	33	"
		"	280	"	GND	"	"	"	"	"	"		OUT		"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	281	"	"	"	"	"	"	"	"			OUT	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	282	"	"	"	"	"	"	"	"	OUT			"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	283	IN	"	"	"	"	"	"	"		OUT		"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	284	GND	"	"	"	"	"	"	"			OUT	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	285	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	286	GND	"	"	"	"	"	"	"				"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
10	Same tests and terminal conditions as subgroup 9, except Tc = 125°C and limits as follows: t_{PHL1} = 2 to 37 ns; t_{PLH1} = 2 to 55 ns; t_{PHL2} and t_{PLH2} = 2 to 55 ns; t_{PHL3} and t_{PLH3} = 2 to 52 ns; t_{PHL4} = 2 to 57 ns; t_{PLH4} = 2 to 52 ns; t_{PHL5} = 2 to 42 ns; t_{PLH5} = 2 to 51 ns; t_{PHL6} and t_{PLH6} = 2 to 55 ns; t_{PHL7} = 2 to 37 ns; t_{PLH7} = 2 to 46 ns; t_{PHL8} = 2 to 96 ns; t_{PLH8} = 2 to 100 ns; t_{PHL9} = 2 to 37 ns; t_{PLH9} = 2 to 48 ns; t_{PHL10} = 2 to 37 ns; t_{PLH10} = 2 to 48 ns; t_{PHL11} and t_{PLH11} = 2 to 64 ns; t_{PHL12} = 2 to 60 ns; t_{PLH12} = 2 to 69 ns; t_{PHL13} = 2 to 48 ns; t_{PLH13} = 2 to 57 ns;																		
11	Same tests, terminal conditions and limits as subgroup 10, except Tc = -55°C																		

See footnotes at end of table. Pins 13 thru 24 on next page.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K, Z	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Case 3 1/	16	17	18	19	20	21	23	24	25	26	27	28				
			Test no.	$\overline{F}$ 3	A = B	$\overline{P}$	Cn+4	$\overline{G}$	$\overline{B}$ 3	$\overline{A}$ 3	$\overline{B}$ 2	$\overline{A}$ 2	$\overline{B}$ 1	$\overline{A}$ 1	V_{CC}		Min	Max	
9	t_{PHL13}	3003	271						GND	GND	GND	GND	GND	GND	5.0 V	$\overline{A}$ 0 to $\overline{F}$ 0	2	38	ns
Tc = 25°C		Fig 4	272						"	GND	"	GND	"	IN	"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	273						"	GND	"	IN	"	GND	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	274	OUT					"	IN	"	GND	"	"	"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	275						"	GND	"	"	"	"	"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	276						"	"	"	"	IN	"	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	277						"	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	278	OUT					IN	"	GND	"	"	"	"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
	t_{PLH13}	"	279						GND	"	"	"	"	"	"	$\overline{A}$ 0 to $\overline{F}$ 0	"	33	"
		"	280						"	"	"	"	"	IN	"	$\overline{A}$ 1 to $\overline{F}$ 1	"	"	"
		"	281						"	"	"	IN	"	GND	"	$\overline{A}$ 2 to $\overline{F}$ 2	"	"	"
		"	282	OUT					"	IN	"	GND	"	"	"	$\overline{A}$ 3 to $\overline{F}$ 3	"	"	"
		"	283						"	GND	"	"	"	"	"	$\overline{B}$ 0 to $\overline{F}$ 0	"	"	"
		"	284						"	"	"	"	IN	"	"	$\overline{B}$ 1 to $\overline{F}$ 1	"	"	"
		"	285						"	"	IN	"	GND	"	"	$\overline{B}$ 2 to $\overline{F}$ 2	"	"	"
		"	286	OUT					IN	"	GND	"	GND	"	"	$\overline{B}$ 3 to $\overline{F}$ 3	"	"	"
10	Same tests and terminal conditions as subgroup 9, except Tc = 125°C and limits as follows: t_{PHL1} = 2 to 37 ns; t_{PLH1} = 2 to 55 ns; t_{PHL8} = 2 to 96 ns; t_{PLH8} = 2 to 100 ns; t_{PHL2} and t_{PLH2} = 2 to 55 ns; t_{PHL9} = 2 to 37 ns; t_{PLH9} = 2 to 48 ns; t_{PHL3} and t_{PLH3} = 2 to 52 ns; t_{PHL10} = 2 to 37 ns; t_{PLH10} = 2 to 48 ns; t_{PHL4} = 2 to 57 ns; t_{PLH4} = 2 to 52 ns; t_{PHL11} and t_{PLH11} = 2 to 64 ns; t_{PHL5} = 2 to 42 ns; t_{PLH5} = 2 to 51 ns; t_{PHL12} = 2 to 60 ns; t_{PLH12} = 2 to 69 ns; t_{PHL6} and t_{PLH6} = 2 to 55 ns; t_{PHL13} = 2 to 48 ns; t_{PLH13} = 2 to 57 ns; t_{PHL7} = 2 to 37 ns; t_{PLH7} = 2 to 46 ns;																	
11	Same tests, terminal conditions and limits as subgroup 10, except Tc = -55°C																		

1/ Tests shall be performed in sequence.
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.7 volts maximum.
3/ Output voltages shall be either: (a) H = 2.5 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H > 1.5 V and L < 1.5 V when using a high speed checker single comparator.
4/ All functional tests (82 thru 120) require a 1 kΩ pull up resistor between pin 14 and V_{CC}.
5/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Test no.	A	B	C	D
I_{IL1}	16	-0.16/-0.40	-0.03/-0.30	-0.16/-0.40	-0.15/-0.38
I_{IL2}	17, 18, 19, 29	-0.48/-1.20	-0.09/-0.90	-0.48/-1.20	-0.45/-1.14
I_{IL3}	21, 22, 23, 24	-0.64/-1.60	-0.72/-1.44	-0.64/-1.60	-0.60/-1.52
I_{IL4}	21, 22, 23, 24	-0.64/-1.60	-0.72/-1.44	-0.64/-1.60	-0.60/-1.52
I_{IL5}	25	-0.80/-2.0	-1.0/-2.0	-0.80/-2.0	-0.75/-1.90
I_{IL6}	26, 27, 28, 29	-0.32/-0.80	-0.09/-0.09	0.32/-0.82	-0.30/-0.76

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements (see 5.1).

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

GND ..... Ground zero voltage potential
V_{IN} ..... Voltage level at an input terminal
V_{IC} ..... Input clamp voltage
I_{IN} ..... Current flowing into an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	54LS181

6.8 Manufacturers' designation. Manufacturers' circuits, which form a part of this specification, are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturer's designator.

Device type	A	B	C	D
	Texas Instruments	Signetics Corporation	Raytheon Company	Motorola Incorporated
01	X	X	X	X

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-1993)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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